

GUTHULA RAJA VENKATESH

Chennai, India | +91 7358383372 | grv.9604@gmail.com | rajavenkatesh.dev | github.com/rajavenkatesh04 | linkedin.com/in/guthularajavenkatesh

SUMMARY

Software Engineer and recent Computer Science graduate with hands-on experience architecting, testing, and scaling full-stack web applications using Java, React, Next.js, and Node.js. Proven ability to deliver high-impact production systems, manage complex relational databases, and integrate AI/LLM technologies. Strong focus on clean code, performance optimization, and building user-centric solutions with real-world adoption.

TECHNICAL SKILLS

Languages: Java, JavaScript, TypeScript, Python, SQL, HTML/CSS

Frameworks: React.js, Next.js, Node.js, Express.js, Tailwind CSS

Databases: PostgreSQL, MySQL, MongoDB, Firebase Firestore, FAISS (vector search)

AI / ML: Retrieval-Augmented Generation (RAG), LangChain, Hugging Face Transformers, LLM integration (Llama, Gemini), spaCy NLP

Tools / Cloud: AWS, Google Cloud Platform, Docker, Git, GitHub, CI/CD, Postman, Agile

PROJECTS

Code-Mixed RAG Chatbot for Tamil Nadu Disaster Recovery | *Python, LangChain, FAISS, Llama-3.3-70B, Gemini*

- Built a production-grade Retrieval-Augmented Generation (RAG) system that answers disaster-relief queries in Tamil, English, and code-mixed Tanglish using Indic-SBERT embeddings and FAISS vector search across 3,000+ document chunks.
- Engineered a modular NLP pipeline with zero-shot intent classification (BART-MNLI), language detection, transliteration-based normalization, and spaCy-based entity extraction for amounts, helplines, and government portals.
- Implemented a dual-LLM architecture (Groq Llama-3.3-70B primary, Gemini 2.0 Flash fallback) for fault tolerance, and validated faithfulness with RAGAS and BERTScore across 150+ test queries.

DeeperWeave — Full-Stack Media Discovery Platform [Live Demo](#) | *Next.js, PostgreSQL (Supabase)*

- Architected a full-stack social platform using Next.js 16 (App Router), React 19, and TypeScript, designing a robust Supabase (PostgreSQL) schema with 15+ tables, Row-Level Security (RLS), and 20+ Zod-validated server actions to securely manage user authentication, role-based access control (RBAC), and a Razorpay-integrated tiered subscription system.
- Engineered a high-performance caching architecture utilizing Next.js ISR, TanStack Query, and a "Lazy Mirror" synchronization pattern to locally cache external TMDB API data, strictly maintaining foreign-key integrity while handling 14+ parallel, region-aware data fetches at 24-hour intervals.
- Implemented a responsive, production-grade UI featuring real-time interactions, debounced full-text search, infinite scrolling, and optimistic updates with instant error-rollback, leveraging Tailwind CSS v4, shadcn/ui, and Framer Motion/GSAP to deliver a polished user experience with integrated SFW content moderation.

Hostels — Hostel Selection Platform [Live Demo](#) | *Next.js, React.js, JavaScript*

- Architected a full-stack hostel discovery platform using Next.js 16, React 19, and TypeScript—currently adopted by SRM College for incoming freshmen, scaling to serve over 22,000+ unique visitors and 120,000+ page views in production.
- Built a highly responsive, mobile-first UI utilizing shadcn/ui, Tailwind CSS v4, and semantic HTML, continuously monitoring user engagement and performance bottlenecks via Vercel Analytics and Microsoft Clarity.
- Engineered an interactive guided chooser with dynamic, URL-synced state management (via [nuqs](#)) and a custom scoring algorithm, optimizing front-end performance and routing to reduce the bounce rate by 17% (down to 31%).

EDUCATION

SRM Institute of Science and Technology, Chennai, India

Bachelor of Technology in Computer Science and Engineering — CGPA: 7.98 / 10.0 (Sep 2022 – May 2026)

Relevant Coursework: Data Structures & Algorithms, Database Management Systems, Object-Oriented Programming, Operating Systems, Computer Networks, Software Engineering

CERTIFICATIONS

- Database Management with MySQL — Meta ([View Certificate](#))
- Programming in Java — NPTEL ([View Certificate](#))
- AWS Machine Learning Foundations — Amazon Web Services ([View Certificate](#))
- Next.js App Router Fundamentals — Vercel ([View Certificate](#))
- React Foundations for Next.js — Vercel ([View Certificate](#))